

Project
of
Automotive Transmission System
(MEN341)

BACHELOR OF TECHNOLOGY

In

MECHANICAL ENGINEERING



Session:- 2025-26

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Project Title

“Wall Wheel Cart
Caddy Design in
Fusion 360 &
FEA in ANSYS”

Theoretical Background and Working Principle of Wall Wheel Cart Caddy

Drawing Overview :

This drawing sheet presents an isometric view of the Wall Wheel Cart Caddy, where all major components are shown with numbered tags for easy identification. On the right side of the drawing, there is a "Parts List" that provides detailed specifications for each item in the assembly, including quantity, part number, and material.

Assembly Components:

Item	Qty	Part Name	Material
1	2	Wall Assembly	Steel
2	2	Cross Tubing Inner	Steel
3	2	Axle Shaft Support	Steel
4	2	8920K172 Low-Carbon Steel Rod	Steel
5	2	8353T51 Side-Mount Hub Wheel	----
6	2	90126A212 Zinc-Plated Washer	Steel
7	2	90177A221 Zinc-Plated Split Ring	Steel

Component Functions:

Wall Assembly (Item 1): These parts form the side walls of the cart, providing essential stability and rigidity to the entire structure.

Cross Tubing Inner (Item 2): These horizontal frame members connect the wall assemblies, making the whole structure strong and integrated.

Axle Shaft Support (Item 3): These components hold the axle rods in place and ensure the wheels are properly aligned.

Low-Carbon Steel Rod (Item 4): This is the main part of the axle, onto which the wheels are mounted, enabling rotational motion.

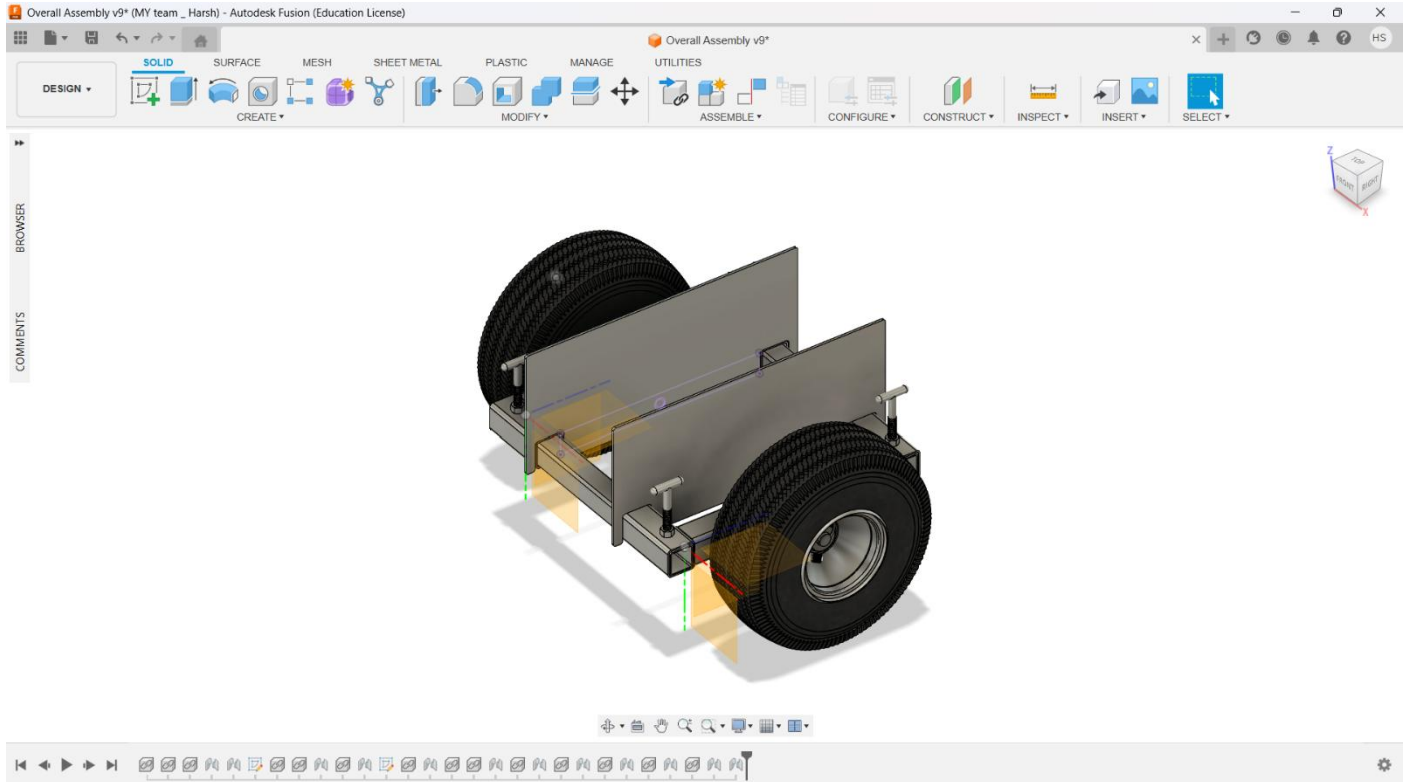
Side-Mount Hub Wheel (Item 5): Pneumatic wheels that deliver smooth movement of the cart over the ground surface.

Zinc-Plated Washer & Split Rings (Items 6 & 7): These are fastening hardware elements used to secure the wheel assembly, preventing unwanted axial movement and separation.

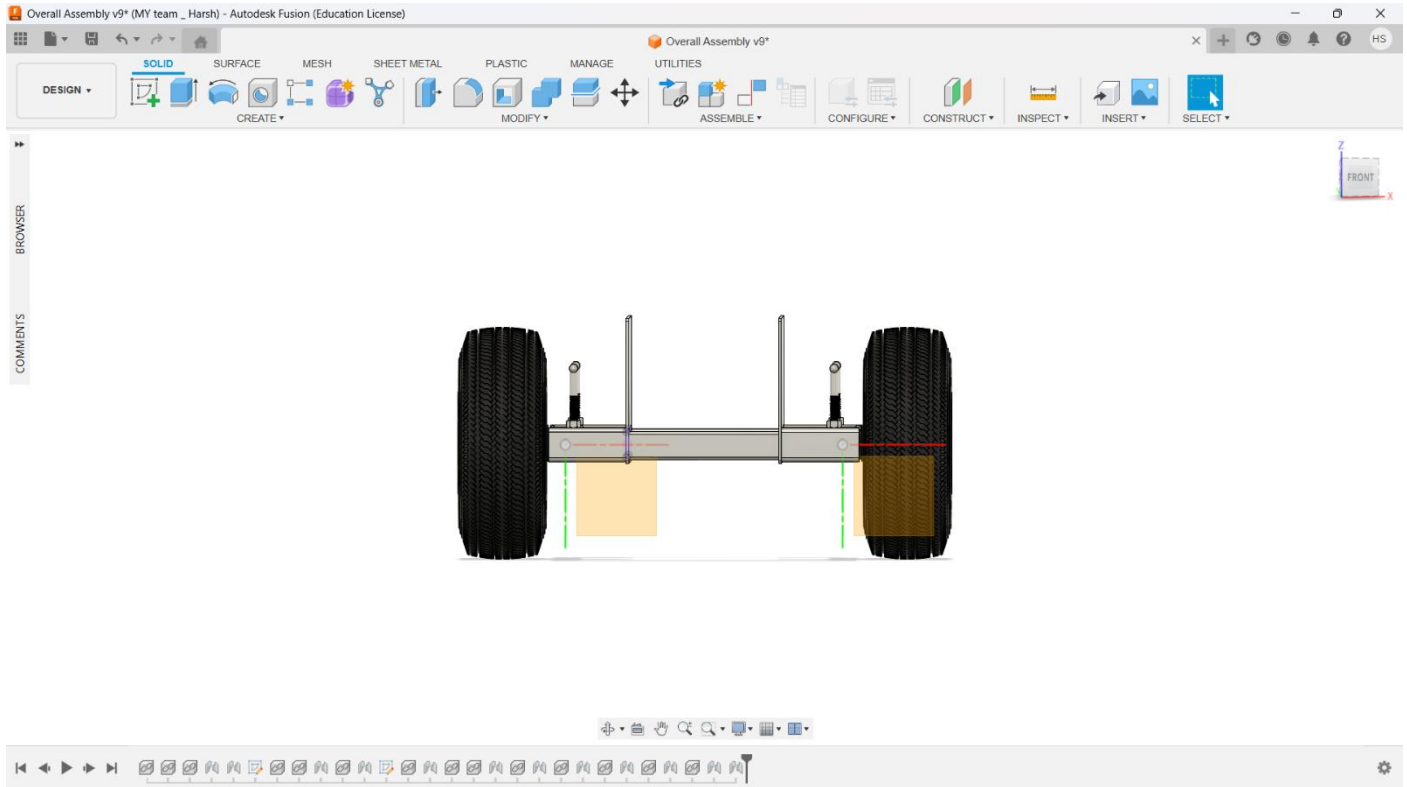
Material Selection Reason:

Most components are made of steel, as steel is strong, durable, and suitable for load-bearing applications. Zinc-plated hardware is chosen for its corrosion resistance, which is important for outdoor use.

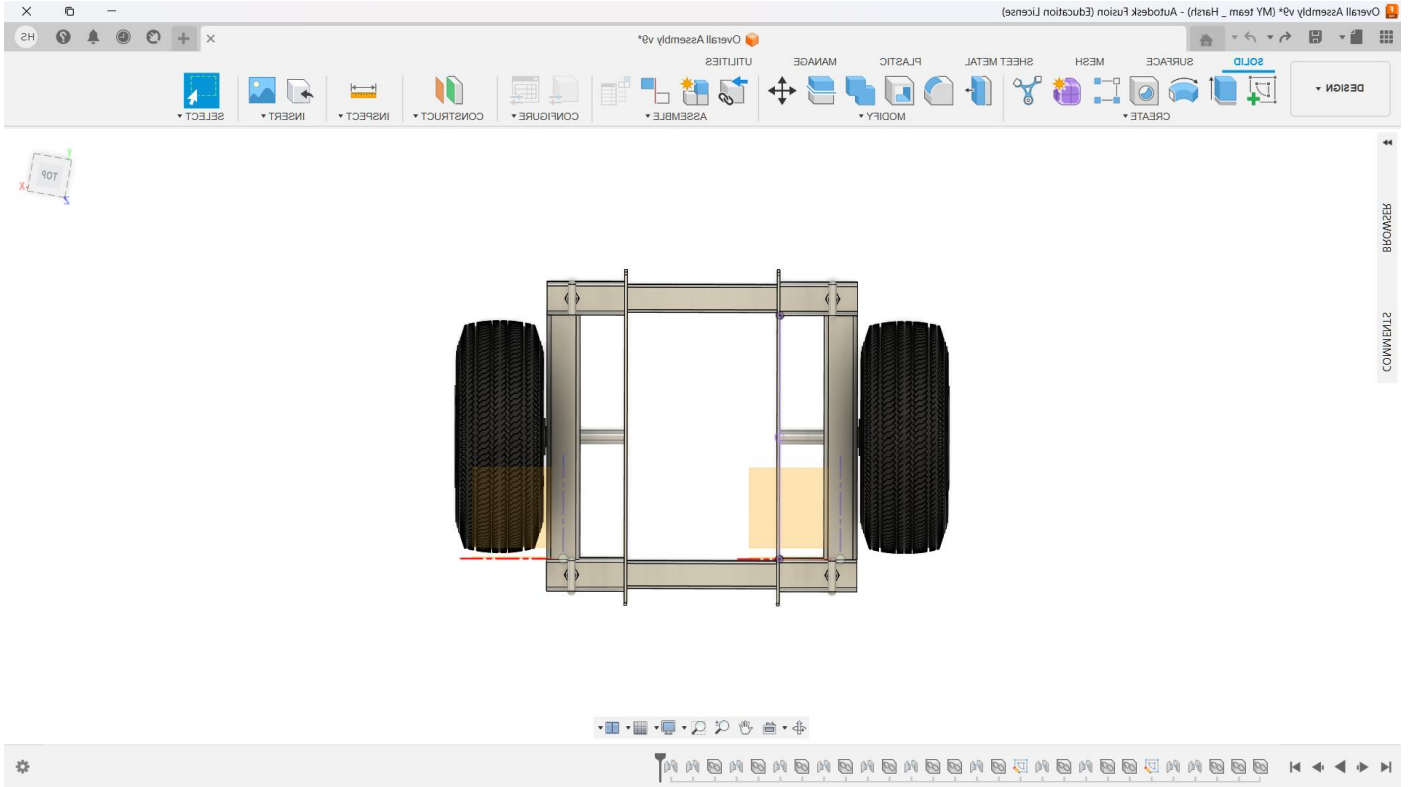
Isometric View



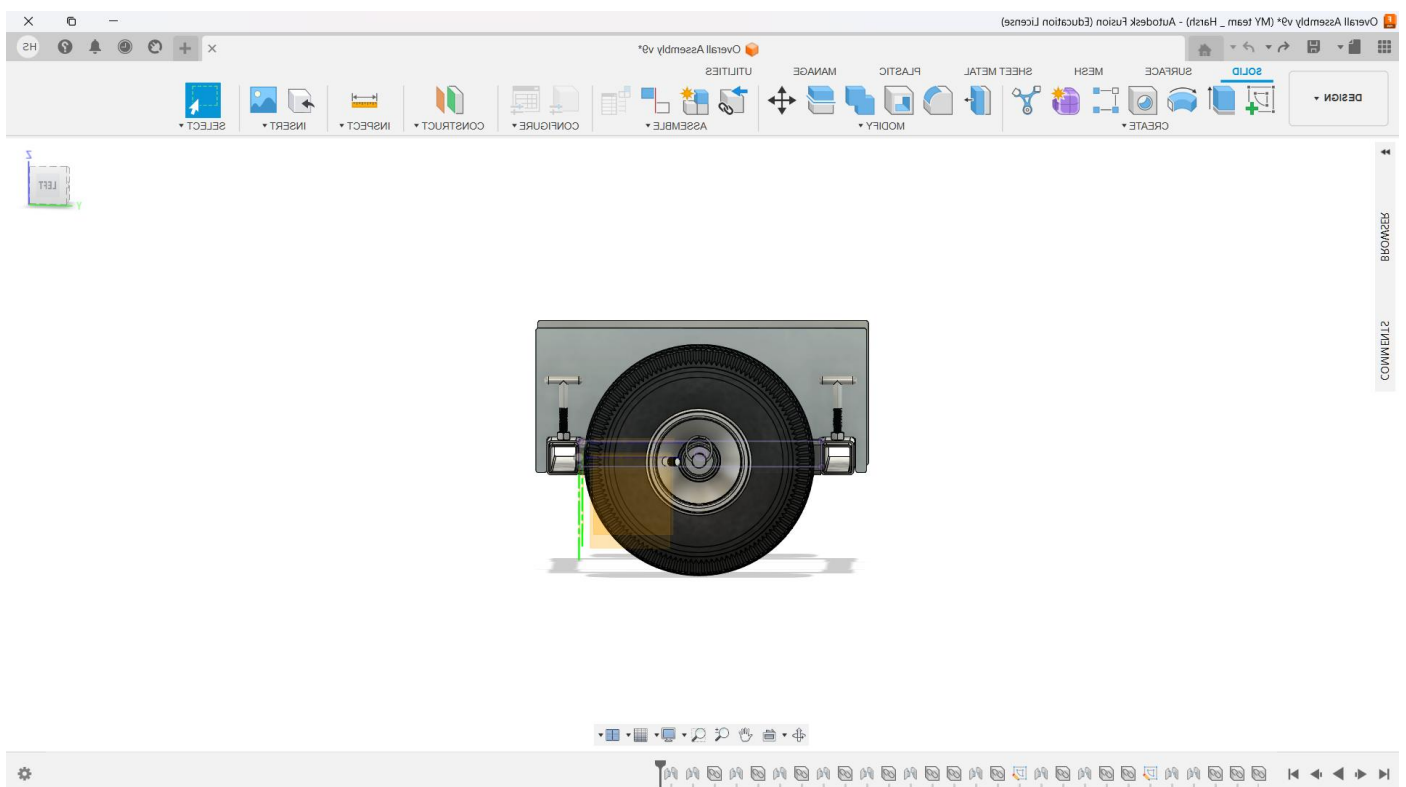
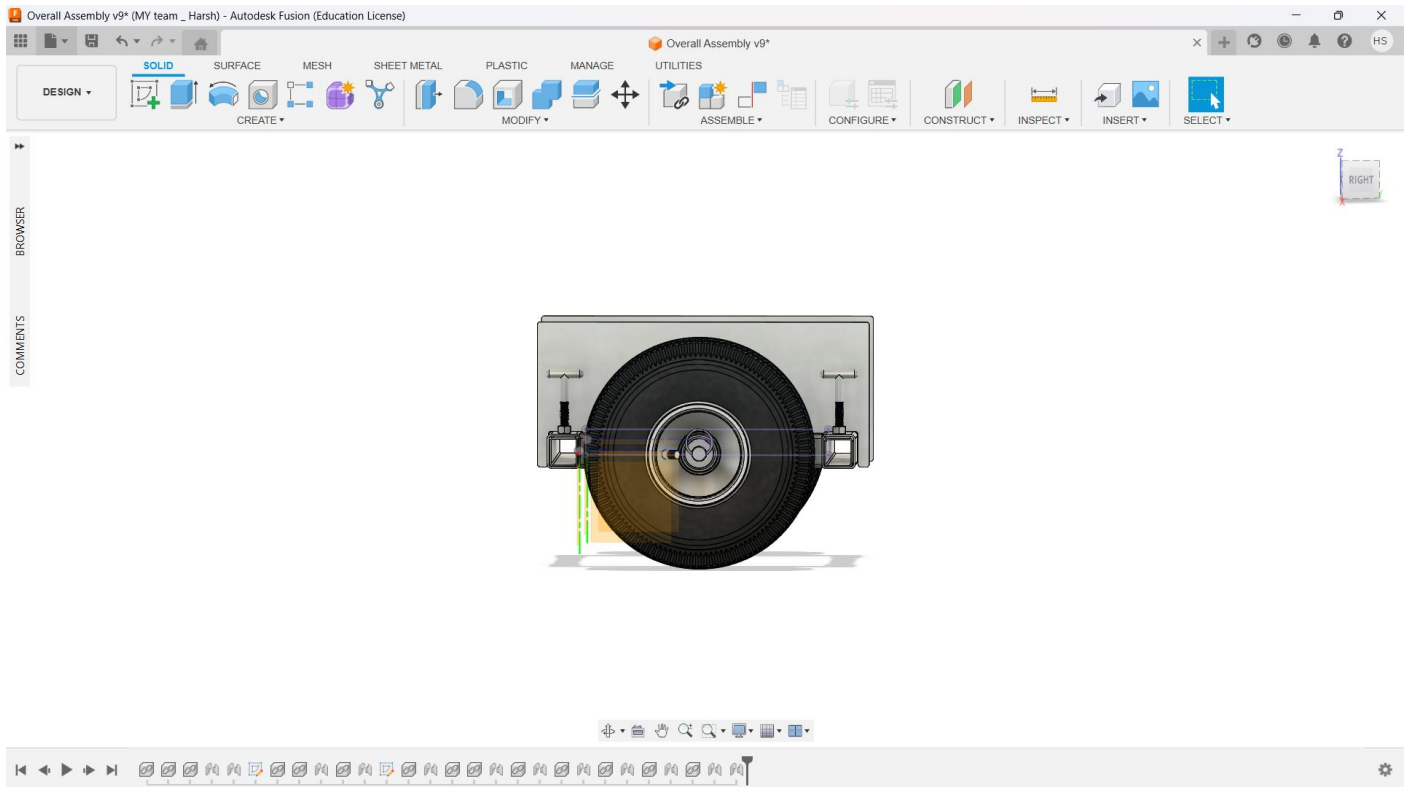
Front View



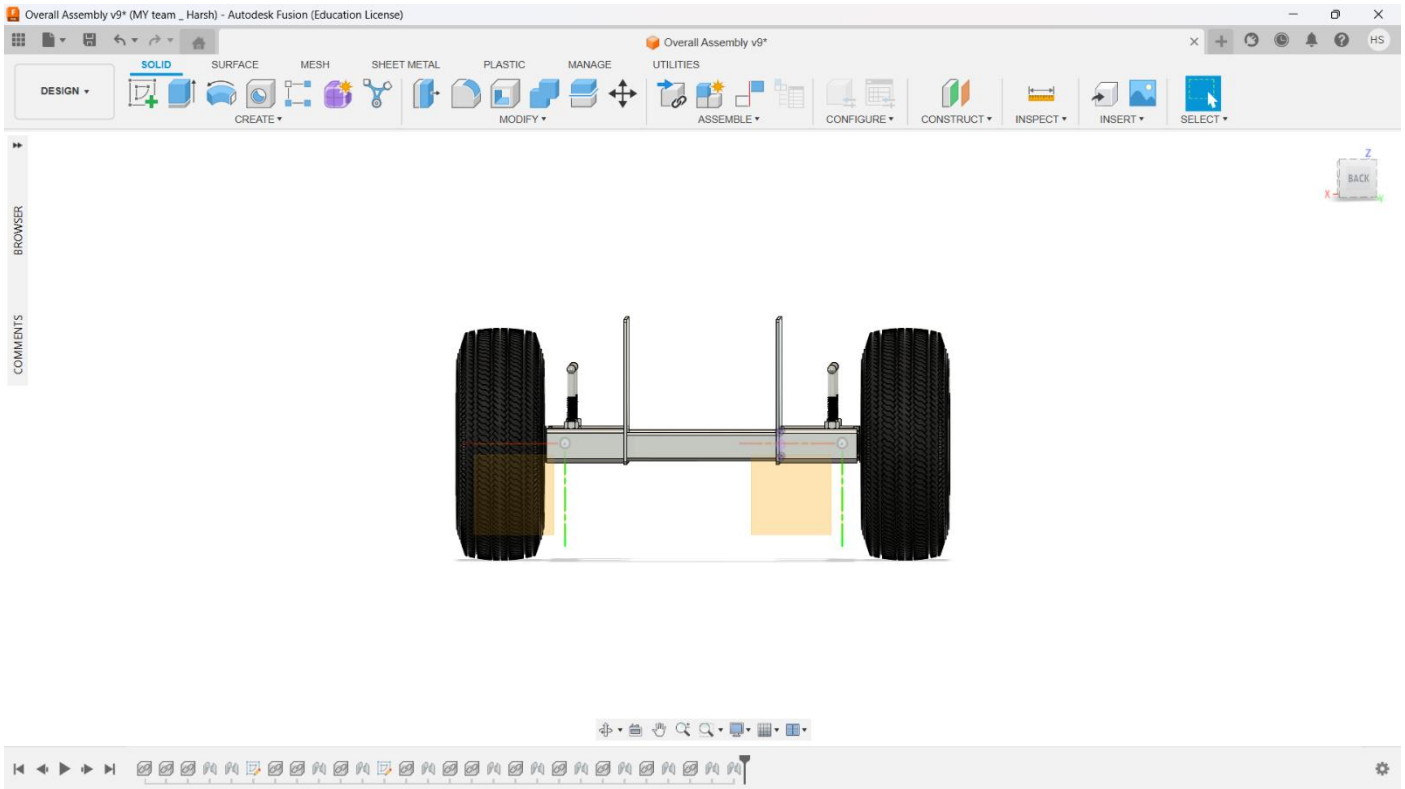
Top View



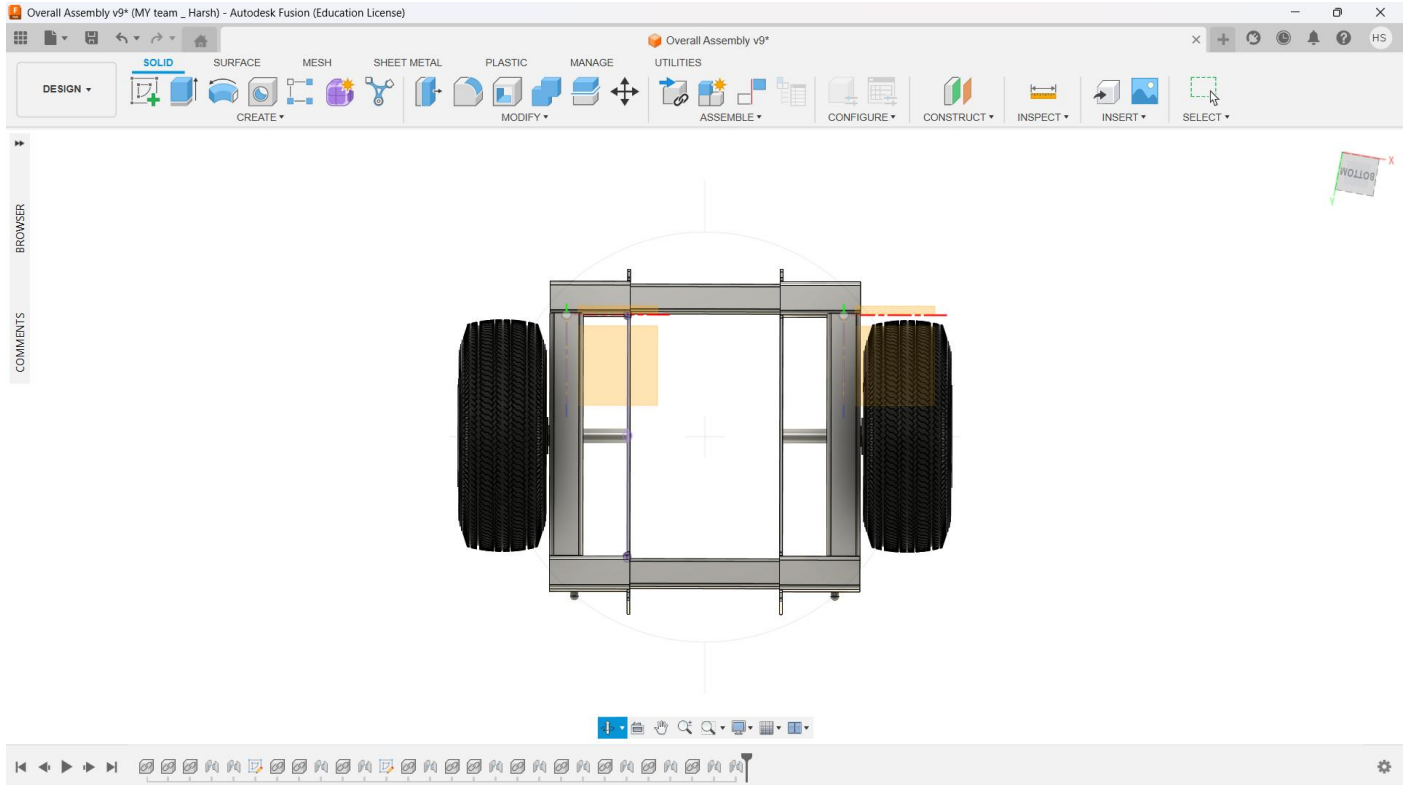
Side View



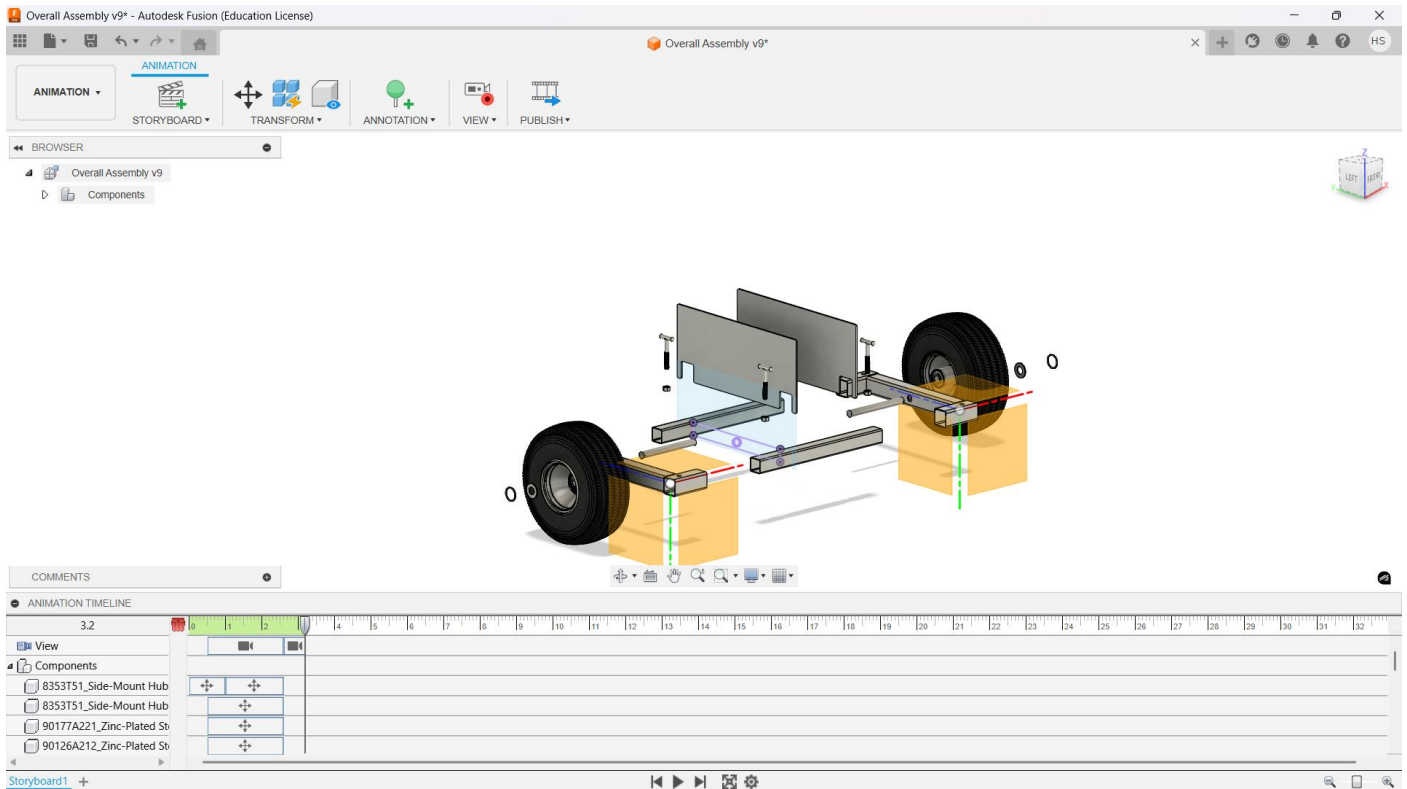
Back View



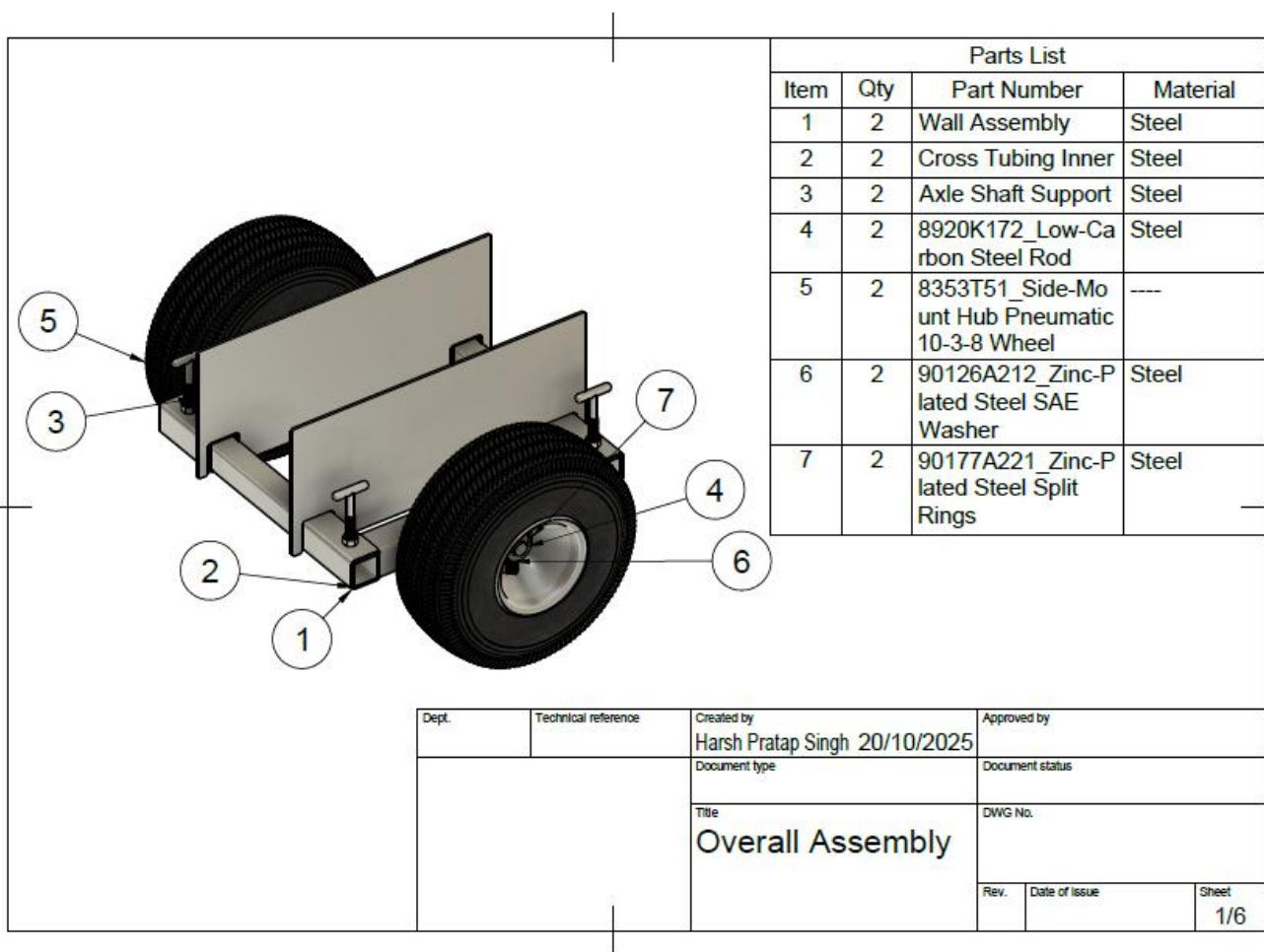
Bottom View



Exploded View

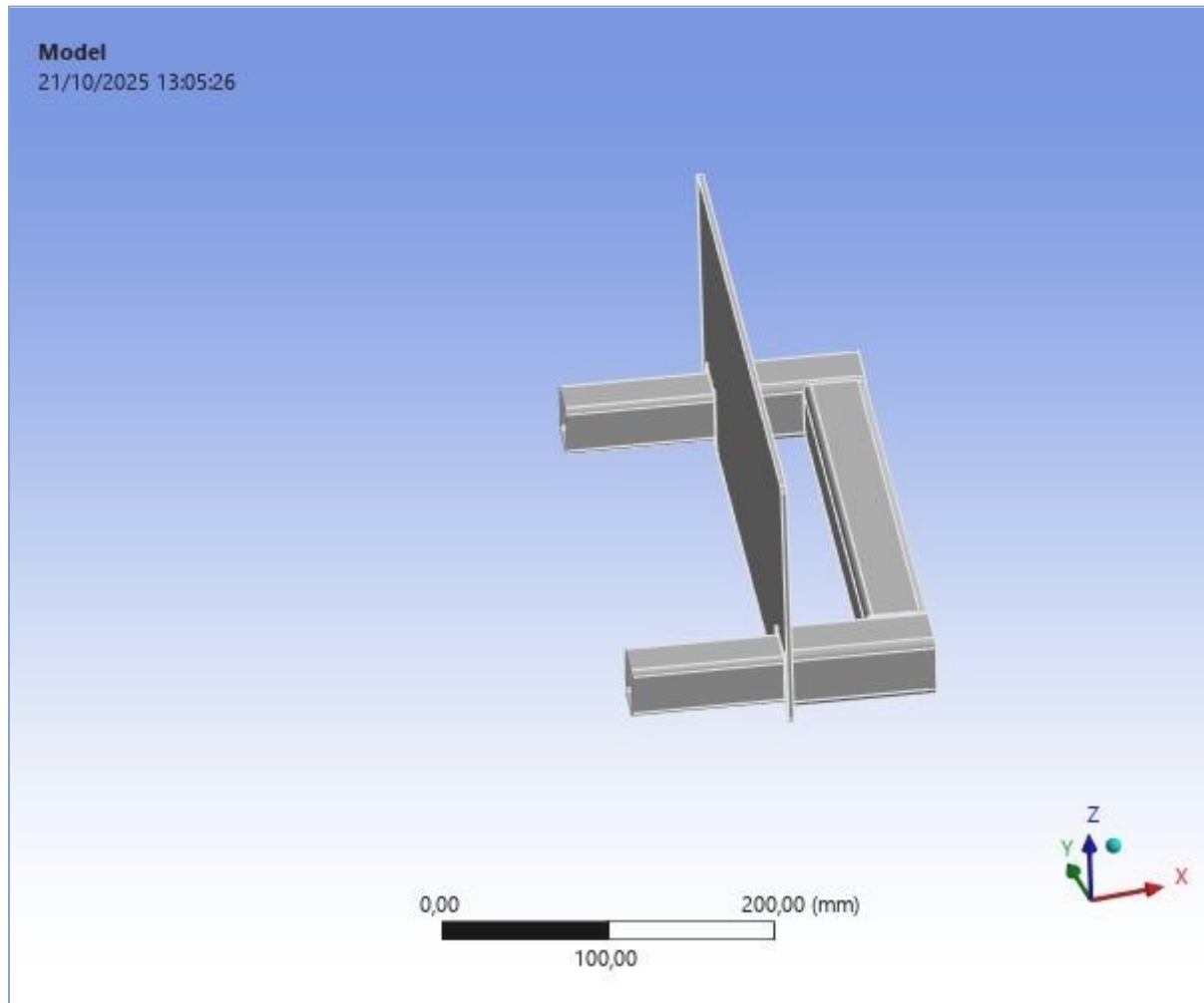


Details of Parts





Wall Wheel Cart Caddy



Contents

1 Units

1 Model (A4)

; Geometry Imports n Geometry Import (A3)

; Geometry n Solid

; Materials

; Coordinate Systems

; Mesh

; Static Structural (A5) n Analysis Settings n

Loads

n Solution (A6) n Solution

Information n Results

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2

Model (A4) > Geometry Imports

Object Name	Geometry Imports
State	Solved

TABLE 3

Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	Geometry Import (A3)
State	Solved
Definition	
Source	C:\Users\Lenovo\AppData\Local\Temp\WB_Lenovo_19504_2\wbnew_files\dp0\SYS\DM\SYS.agdb
Type	DesignModeler
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Advanced Geometry Options	
Compare Parts On Update	No
Analysis Type	3-D

Geometry

TABLE 4

Model (A4) > Geometry

Object Name	Geometry
State	Fully Defined
Definition	
Source	C:\Users\Lenovo\AppData\Local\Temp\WB_Lenovo_19504_2\wbnew_files\dp0\SYS\DM\SYS.agdb
Type	DesignModeler
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	411,44 mm

Length Y	409,09 mm
Length Z	165, mm
Properties	
Volume	7,6716e+005 mm ³
Mass	6,0222 kg
Scale Factor Value	1,
Statistics	
Bodies	1
Active Bodies	1
Nodes	106750
Elements	52318
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	Solid
State	Meshed
Graphics Properties	
Visible	Yes

Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Structural Steel
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	411,44 mm
Length Y	409,09 mm
Length Z	165, mm
Properties	
Volume	7,6716e+005 mm ³
Mass	6,0222 kg
Centroid X	-128,84 mm
Centroid Y	215,48 mm
Centroid Z	52,692 mm
Moment of Inertia Ip1	1,0087e+005 kg·mm ²
Moment of Inertia Ip2	92413 kg·mm ²
Moment of Inertia Ip3	1,665e+005 kg·mm ²
Statistics	
Nodes	106750
Elements	52318
Mesh Metric	None

TABLE 6
Model (A4) > Materials

Object Name	Materials
State	Fully Defined
Statistics	
Materials	1
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	Global Coordinate System
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0,
Origin	

Origin X	0, mm
Origin Y	0, mm
Origin Z	0, mm
Directional Vectors	
X Axis Data	[1, 0, 0,]
Y Axis Data	[0, 1, 0,]
Z Axis Data	[0, 0, 1,]
Transfer Properties	
Source	
Read Only	No

Mesh

TABLE 8
Model (A4) > Mesh

Object Name	Mesh
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	5, mm
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	603,21 mm
Average Surface Area	3005,8 mm ²
Minimum Edge Length	0,11199 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5,e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0,272
Maximum Layers	5

Growth Rate	1,2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	106750
Elements	52318
Show Detailed Statistics	No

Static Structural (A5)

TABLE 9
Model (A4) > Analysis

Object Name	Static Structural (A5)
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22, °C
Generate Input Only	No

TABLE 10
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	Analysis Settings
State	Fully Defined
Step Controls	
Number Of Steps	1,
Current Step Number	1,
Step End Time	1, s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off

Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	D:\Fusion Model\Wall Wheel Cart Caddy_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No

Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	mmm

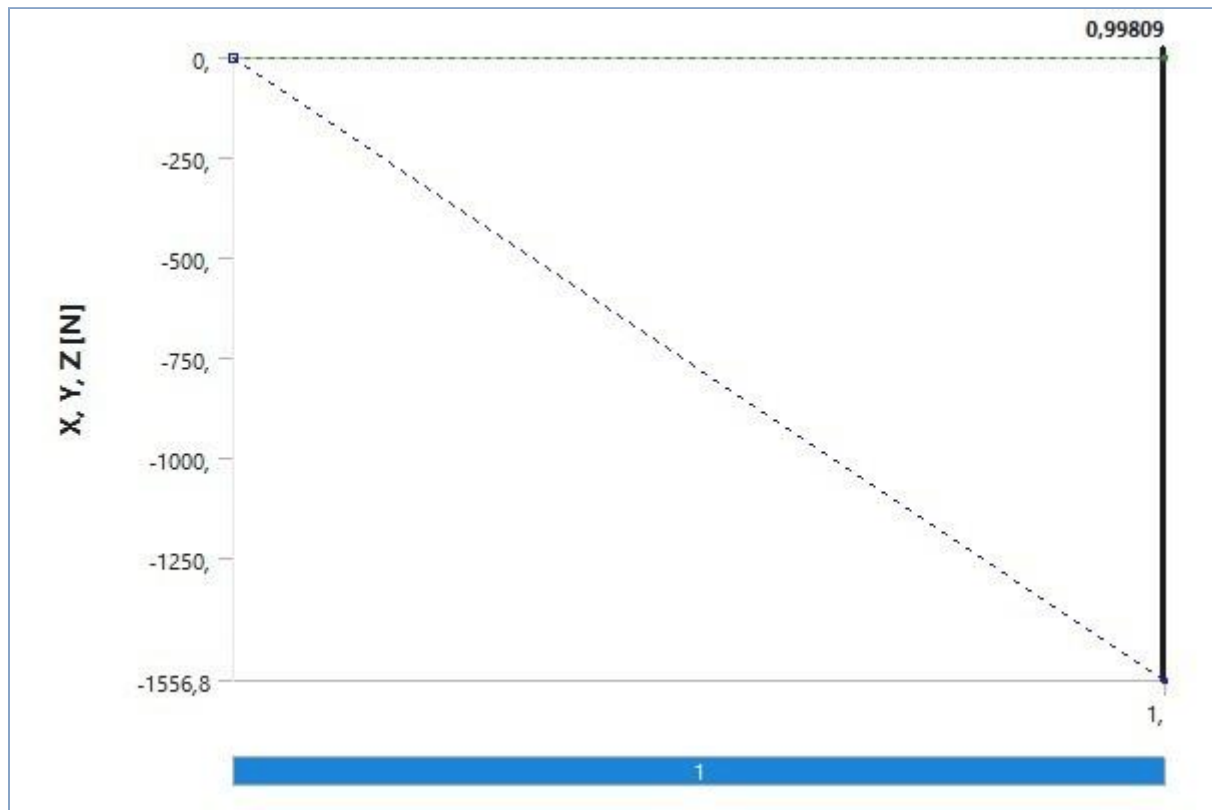
TABLE 11

Model (A4) > Static Structural (A5) > Loads

Object Name	Fixed Support	Force
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	174 Faces	8 Faces
Definition		
Type	Fixed Support	Force
Suppressed	No	
Define By		Components
Applied By		Surface Effect
Coordinate System		Global Coordinate System
X Component		0, N (ramped)
Y Component		0, N (ramped)
Z Component		-1556,8 N (ramped)

FIGURE 1

Model (A4) > Static Structural (A5) > Force



Solution (A6)

TABLE 12
Model (A4) > Static Structural (A5) > Solution

Object Name	Solution (A6)
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1,
Refinement Depth	2,
Information	
Status	Done
MAPDL Elapsed Time	24, s
MAPDL Memory Used	922, MB
MAPDL Result File Size	35,625 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 13
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	Solution Information
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2,5 s

Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 14

Model (A4) > Static Structural (A5) > Solution (A6) > Results

Object Name	Equivalent Stress	Total Deformation
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Equivalent (von-Mises) Stress	Total Deformation
By	Time	
Display Time	Last	
Separate Data by Entity	No	
Calculate Time History	Yes	
Identifier		
Suppressed	No	
Integration Point Results		
Display Option	Averaged	
Average Across Bodies	No	
Results		
Minimum	0, MPa	0, mm
Maximum	0,12777 MPa	1,5079e-006 mm
Average	6,4693e-003 MPa	4,8671e-008 mm
Minimum Occurs On	Solid	
Maximum Occurs On	Solid	
Information		
Time	1, s	
Load Step	1	
Substep	1	
Iteration Number	1	

FIGURE 2

Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

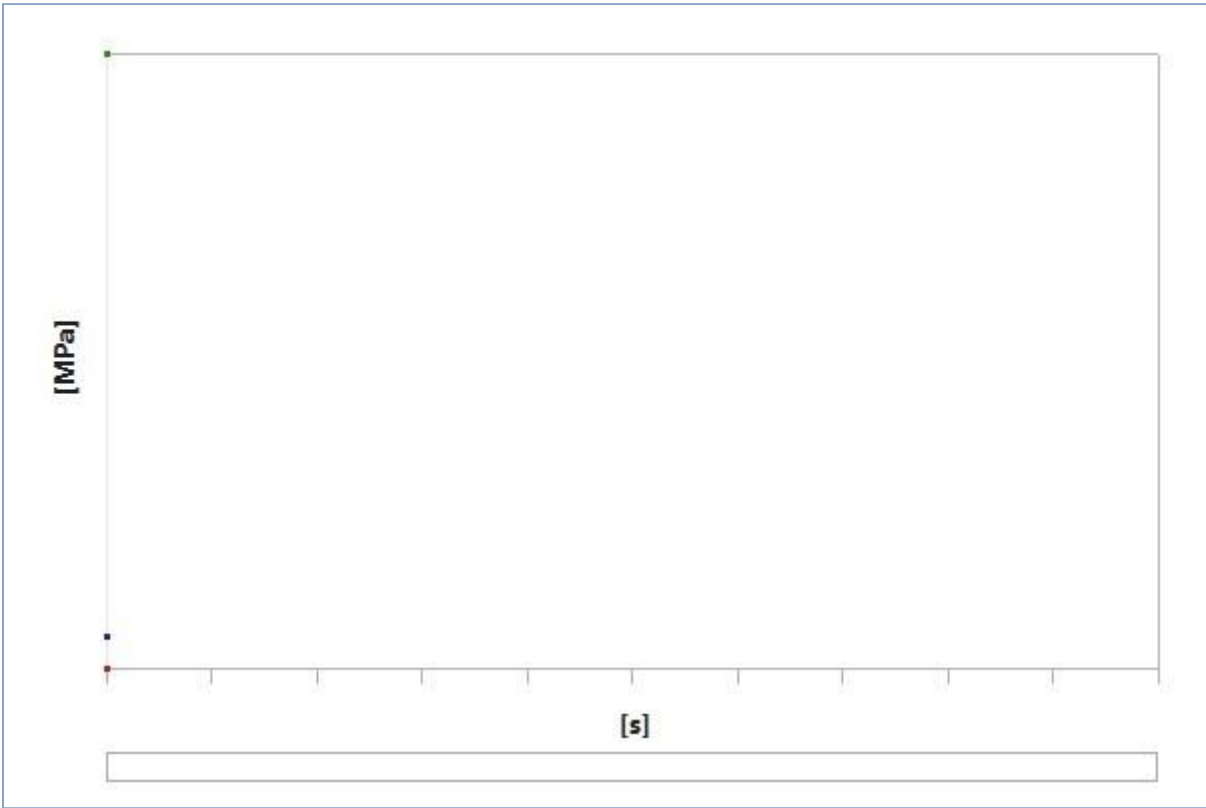


TABLE 15
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1,	0,	0,12777	6,4693e-003

FIGURE 3
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress > Figure

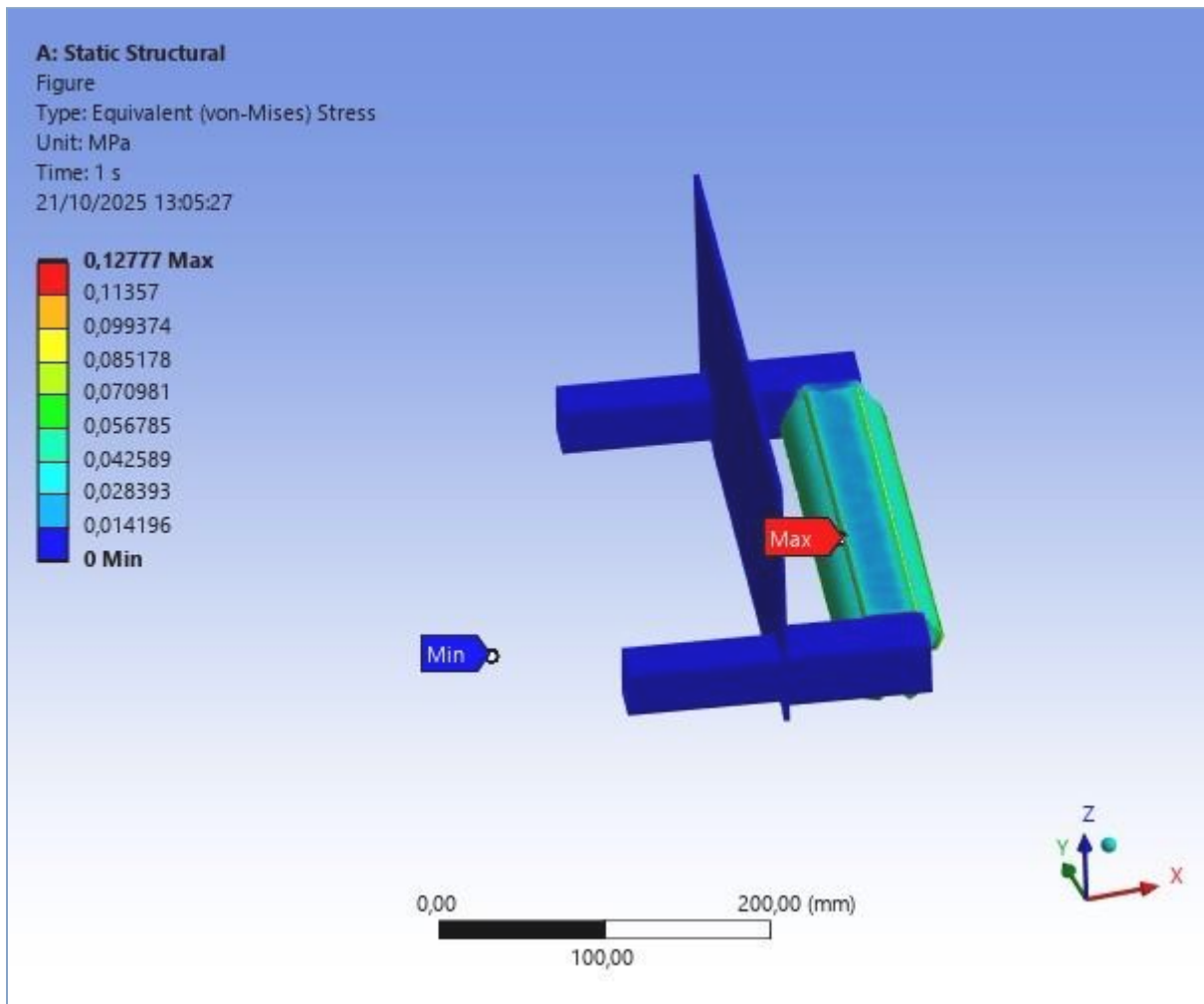


FIGURE 4
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

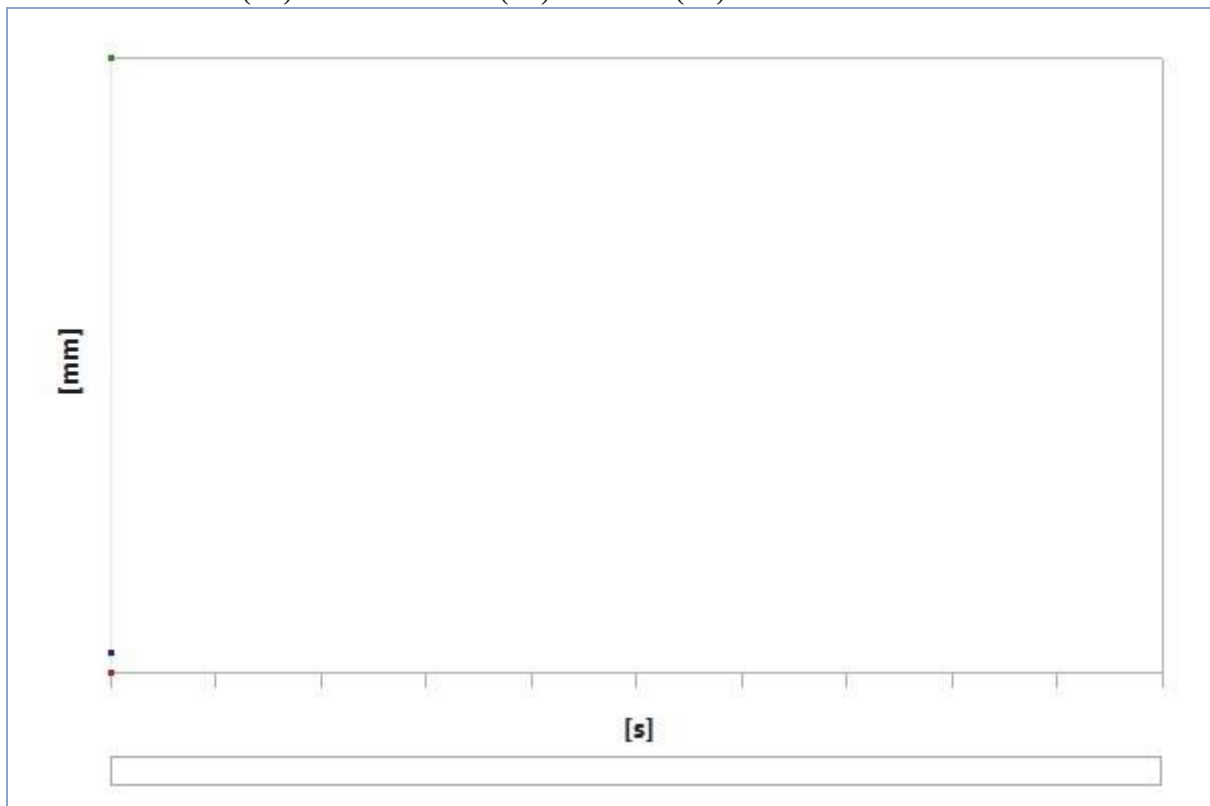


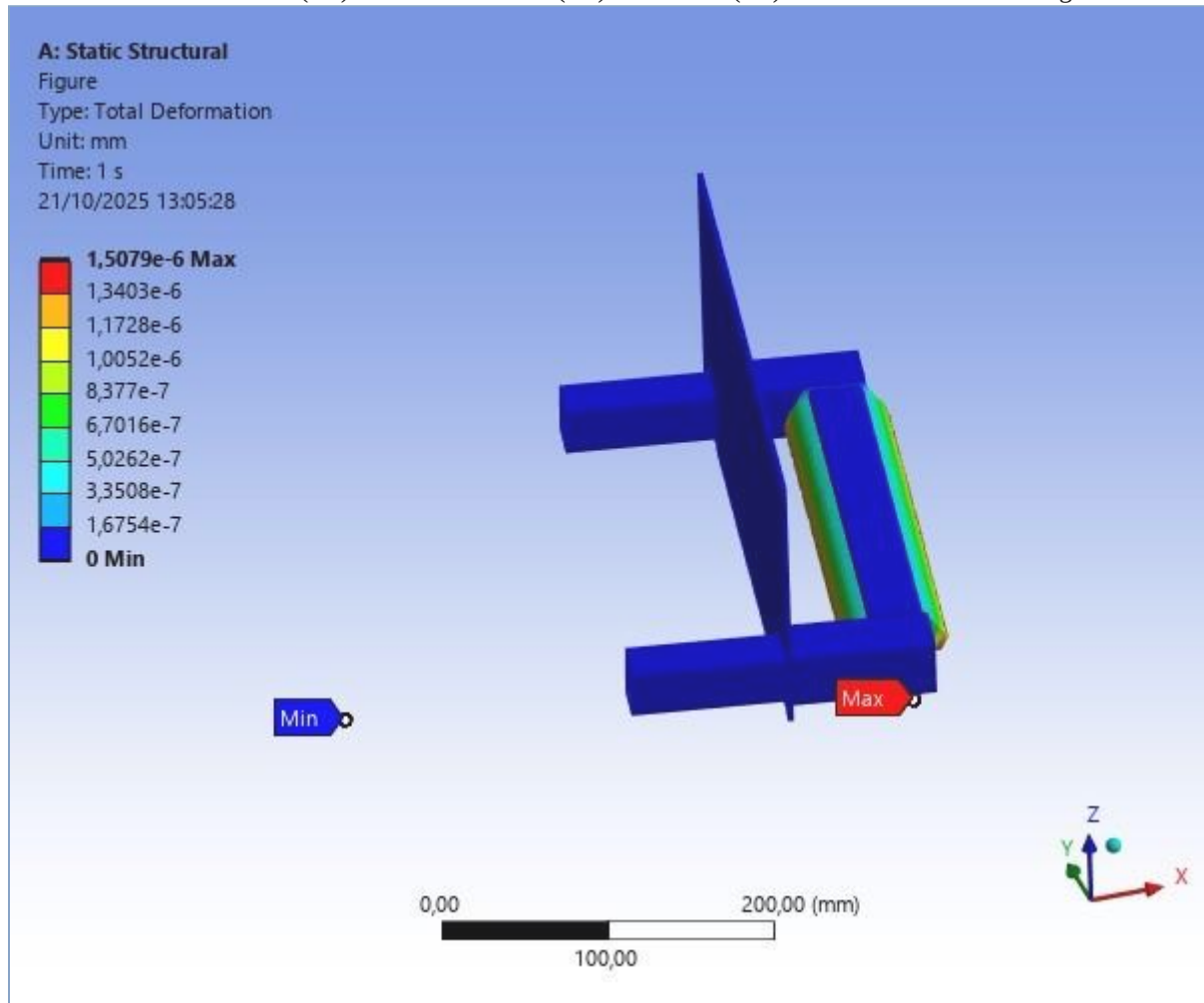
TABLE 16

Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
1,	0,	1,5079e-006	4,8671e-008

FIGURE 5

Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation > Figure



Material Data

Structural Steel

TABLE 17

Structural Steel > Constants

Density	7,85e-006 kg mm ⁻³
Coefficient of Thermal Expansion	1,2e-005 K ⁻¹
Specific Heat	4,34e+005 mJ kg ⁻¹ K ⁻¹
Thermal Conductivity	6,05e-002 W mm ⁻¹ K ⁻¹
Resistivity	1,7e-004 ohm mm

TABLE 18

Structural Steel > Color

Red	Green	Blue
132,	139,	179,

TABLE 19

Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0,

TABLE 20

Structural Steel > Compressive Yield Strength

Compressive Yield Strength MPa
250,

TABLE 21

Structural Steel > Tensile Yield Strength

Tensile Yield Strength MPa
250,

TABLE 22

Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
460,

TABLE 23

Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature K
295,15

TABLE 24

Structural Steel > S-N Curve

Alternating Stress MPa	Cycles	Mean Stress MPa
3999,	10,	0,
2827,	20,	0,
1896,	50,	0,
1413,	100,	0,
1069,	200,	0,
441,	2000,	0,
262,	10000	0,
214,	20000	0,
138,	1,e+005	0,
114,	2,e+005	0,
86,2	1,e+006	0,

TABLE 25

Structural Steel > Strain-Life Parameters

Strength Coefficient MPa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient MPa	Cyclic Strain Hardening Exponent
920,	-0,106	0,213	-0,47	1000,	0,2

TABLE 26

Structural Steel > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
2,e+005	0,3	1,6667e+005	76923	

TABLE 27

Structural Steel > Isotropic Relative Permeability

Relative Permeability
10000